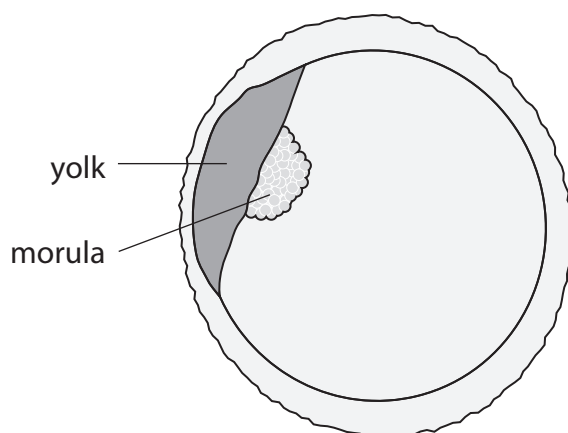


7 Cells can divide by mitosis.

The diagram shows a morula beginning to develop on the yolk of a fertilised fish egg cell.



(a) (i) Which of the following describes a morula?

(1)

- ☐ **A** a hollow ball of pluripotent stem cells
- ☐ **B** a hollow ball of totipotent stem cells
- ☐ **C** a solid ball of pluripotent cells
- ☐ **D** a solid ball of totipotent cells

(ii) Suggest **one** function of the yolk in the fertilised fish egg cell.

(1)

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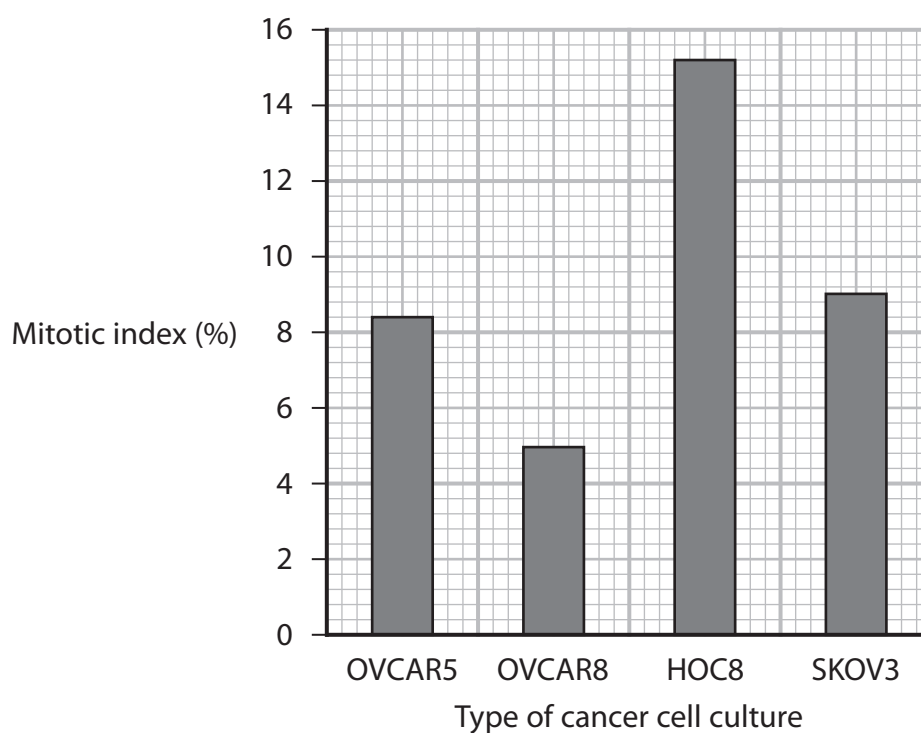


- (5)

[illegible]

(c) The mitotic index of four types of cancer cell culture was investigated.

The graph shows the results of this investigation.



(i) Two thousand OVCAR8 cells were studied.

Calculate how many OVCAR8 cells were in interphase.

(2)

Answer



(ii) Which cancer cell culture has the highest proportion of cells undergoing nuclear division?

(1)

- ☐ **A** HOC8
- ☐ **B** OVCAR5
- ☐ **C** OVCAR8
- ☐ **D** SKOV3

(iii) Explain why some people think that research into cancer treatments should not use embryonic stem cells.

(2)

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(Total for Question 7 = 12 marks)

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